

EGERTON



UNIVERSITY

UNIVERSITY EXAMINATIONS

REGULAR -NJORO CAMPUS

SECOND SEMESTER, 2020/2021 ACADEMIC YEAR

**THIRD YEAR EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN
ENVIRONMENTAL SCIENCE**

ENSC 333: ENVIRONMENTAL HEALTH

STREAM: B.Sc. (ENSC)

TIME: 2 HRS

EXAMINATION SESSION: DECEMBER

YEAR: 2021

INSTRUCTIONS:

- (i) Attempt all questions in Section A and any TWO in Section B
- (ii) Do not write on the question paper
- (iii) Illustrate your answer with suitable diagrams and formulas where appropriate

SECTION A (40 MARKS)

Question One

Write short notes on the following terms as used in Environmental Health.

- a) Fundamental principle of toxicology **(3 Marks)**
- b) Limitation to threshold dose-response data **(4 Marks)**
- c) Individual susceptibility **(3 Marks)**

Question Two

Differentiate between the following terms as used in environmental health.

- a) Systemic and local effects **(3 Marks)**
- b) Bio-accumulation and bio-magnifications **(4 Marks)**

c) Waterborne and water-based diseases

(3 Marks)

Question Three

Describe the processes that modify, remove and sequester chemicals from the environment.

(10 Marks)

Question Four

Persons, animals and plants are exposed to more than one chemical at a time. In the light of this statement, discuss the effect of exposure to multiple chemicals.

(10 Marks)

SECTION B (30 MARKS)

Question Five

All cells possess an inherent rate of cellular division, one in which the rate of normal death cell is matched by the formation of new cells. Discuss in detail how this proliferation of cells develops to form cancer.

(15 Marks)

Question Six

- a) Explain the primary environmental and health risks associated with genetically modified food
- b) Describe in detail the components of an effective sanitation program aimed at preventing foodborne illnesses associated with biological agents and chemical intoxication

(3 Marks)

(12 Marks)

Question Seven

Dose-response assessment is an important step in risk assessment with respect to environmental toxicants. Discuss the various steps to be considered in dose-response assessment.

(15 Marks)
